

9.3. Minimizing Induced Drag, the Elliptic Wing

From Eq. (9.2.19), we see that the minimum induced drag, for a given aspect ratio and lift coefficient, will occur when the span efficiency factor is one. From Eq. (9.2.20) we see that this requires all of the Fourier coefficients except B_1 to be equal to zero. If this is the case, the circulation distribution given by Eq. (9.2.3) becomes

$$\Gamma(\theta) = 2bV_\infty B_1 \sin(\theta) \quad (9.3.1)$$

This is the circulation distribution that will result in minimum induced drag for a given aspect ratio and lift coefficient. Using Eq. (9.3.1) in Eq. (9.2.8), Prandtl's lifting-line equation becomes

$$B_1 \sin(\theta) \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{1}{\sin(\theta)} \right] = \alpha(\theta) - \alpha_{L0}(\theta) \quad (9.3.2)$$

In the previous section, we assumed that all parameters associated with the wing design were known and we used Prandtl's lifting-line equation to solve for the circulation distribution. Here, we shall treat the circulation distribution as known and search for a wing design that will satisfy Eq. (9.3.2). Rearranging Eq. (9.3.2), we can write

$$B_1 = \frac{\alpha(\theta) - \alpha_{L0}(\theta)}{\frac{4b \sin(\theta)}{\widetilde{C}_{L,\alpha} c(\theta)} + 1} \quad (9.3.3)$$

The left-hand side of Eq. (9.3.3) is independent of θ . Thus, if our wing is to produce the optimum circulation distribution given by Eq. (9.3.1), it must be designed such that the right-hand side of Eq. (9.3.3) is also independent of θ . The numerator in the term on the right-hand side of this equation will be independent of θ , if the wing is designed to have no geometric twist and no aerodynamic twist. The denominator in this same term will be independent of θ , if the chord length is proportional to $\sin(\theta)$. Thus, for minimum induced drag, we require

$$c(\theta) = c_r \sin(\theta) \quad (9.3.4)$$

where c_r is the root chord length. The total planform area of the wing is

$$S_p = \int_{z=-b/2}^{b/2} c(z) dz \quad (9.3.5)$$

Using Eqs. (9.2.1), (9.2.2) and (9.3.4) in Eq. (9.3.5) gives

$$S_p = \frac{1}{2} b c_r \int_{\theta=0}^{\pi} \sin^2(\theta) d\theta = \frac{\pi b c_r}{4} \quad (9.3.6)$$

From Eq. (9.3.6) we can relate the root chord length to the wing span and the aspect ratio,

$$c_r = \frac{4S_p}{\pi b} = \frac{4b}{\pi b^2 / S_p} = \frac{4b}{\pi R_A} \quad (9.3.7)$$

Using Eq. (9.3.7) in Eq. (9.3.4), we have

$$c(\theta) = \frac{4b}{\pi R_A} \sin(\theta) \quad (9.3.8)$$

Now, for this wing with no geometric or aerodynamic twist and a planform shape that is specified by Eq. (9.3.8), we can evaluate B_1 . Using Eq. (9.3.8) in Eq. (9.3.3) results in

$$B_1 = \frac{\alpha - \alpha_{L=0}}{\pi R_A / \tilde{C}_{L,\alpha} + 1} = \frac{\tilde{C}_{L,\alpha}(\alpha - \alpha_{L0})}{\pi R_A + \tilde{C}_{L,\alpha}} \quad (9.3.9)$$

From Eq. (9.2.1), we can write

$$\cos(\theta) = -\frac{2z}{b} \quad (9.3.10)$$

or

$$\sin(\theta) = \sqrt{1 - (2z/b)^2} \quad (9.3.11)$$

Using Eq. (9.3.11) in Eq. (9.3.8), we find that we can minimize the induced drag with a wing that has no geometric twist, no aerodynamic twist, and a chord length that varies in the spanwise direction according to the equation

$$c(z) = \frac{4b}{\pi R_A} \sqrt{1 - (2z/b)^2} \quad (9.3.12)$$

This wing has a chord length that varies elliptically with the distance z measured along the span, as is shown in Fig. 9.3.1. Hence, this wing is called an elliptic wing. While the chord length of the wing shown in Fig. 9.3.1 varies elliptically with the spanwise coordinate, z , the planform shape does not look like a conventional ellipse. This is because the theoretical aerodynamic center of an airfoil is the quarter-chord. Thus, an unswept elliptic wing has a straight quarter-chord line. This gives the leading edge less curvature and the trailing edge more curvature than a conventional ellipse, which has a straight half-chord line. Several aircraft have been designed with elliptic wings. One of the best known is the famous British Spitfire shown in Fig. 9.3.2.

Using Eqs. (9.3.9) and (9.3.11) in Eq. (9.3.1), we see that the circulation distribution for an elliptic wing is also elliptic,

$$\Gamma(z) = \frac{2bV_\infty \tilde{C}_{L,\alpha}(\alpha - \alpha_{L0})}{\pi R_A + \tilde{C}_{L,\alpha}} \sqrt{1 - (2z/b)^2} \quad (9.3.13)$$

Since the local section lift is proportional to the local section circulation, the local section lift also varies elliptically with the spanwise coordinate. Figure 9.3.3 shows the local section lift distribution for an unswept elliptic wing with an aspect ratio of 6.

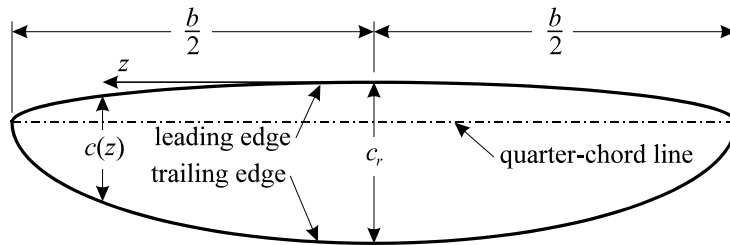


Figure 9.3.1. The planform shape of an unswept elliptic wing with an aspect ratio of 6.

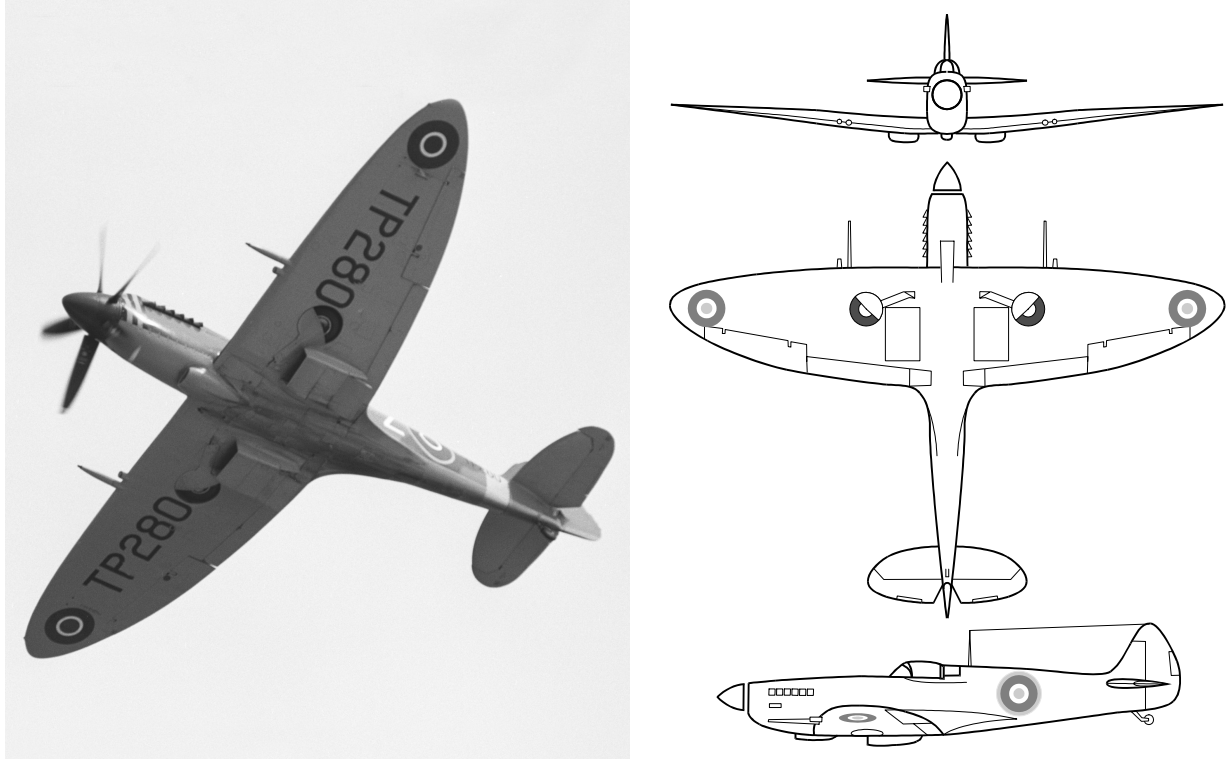


Figure 9.3.2. The elliptic wing used on the famous British Spitfire (photograph by Barry Santana).

Now that we know that an elliptic wing produces the minimum induced drag for a given lift coefficient and aspect ratio, let us examine some other properties of the elliptic wing. The lift coefficient for an elliptic wing is found by using Eq. (9.3.9) in Eq. (9.2.14),

$$C_L = \pi R_A B_1 = \frac{\pi R_A \tilde{C}_{L,\alpha}(\alpha - \alpha_{L0})}{\pi R_A + \tilde{C}_{L,\alpha}} = C_{L,\alpha}(\alpha - \alpha_{L0}) \quad (9.3.14)$$

where the lift slope for the wing, $C_{L,\alpha}$, is related to the section lift slope, $\tilde{C}_{L,\alpha}$, by

$$C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{1 + \tilde{C}_{L,\alpha} / \pi R_A} \quad (9.3.15)$$

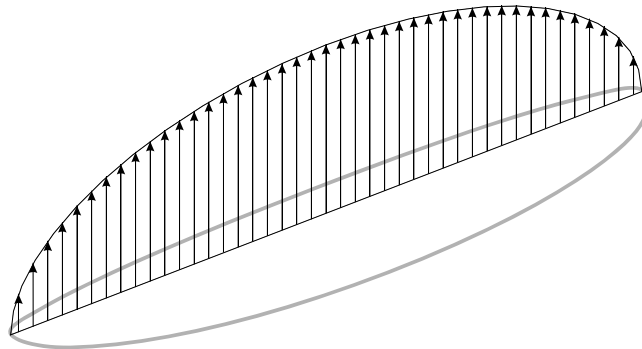


Figure 9.3.3. The local section lift distribution for an unswept elliptic wing with an aspect ratio of 6.

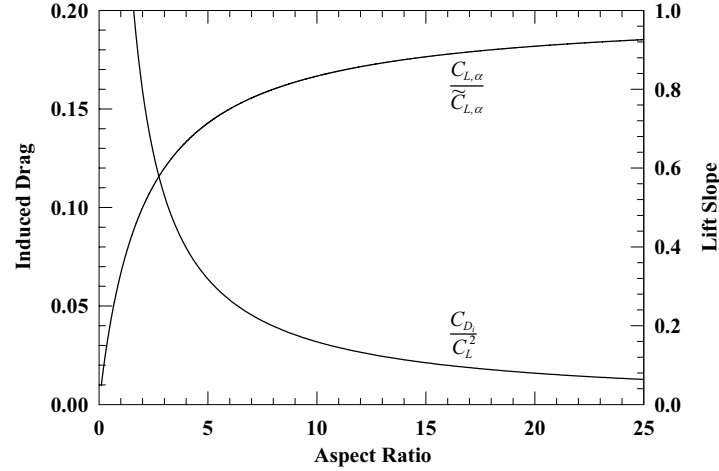


Figure 9.3.4. The effect of aspect ratio on lift and induced drag for the elliptic wing.

Equation (9.3.14) shows that, at a given angle of attack, the lift coefficient for a finite elliptic wing is less than the section lift coefficient for the airfoil from which the wing was generated. However, as the aspect ratio becomes large, the lift coefficient for the wing approaches that of the airfoil section. Figure 9.3.4 shows how the ratio of lift slope to section lift slope varies with aspect ratio for an elliptic wing.

The induced drag coefficient for the finite elliptic wing is obtained from Eq. (9.2.19) with the span efficiency set to unity, which simply restates the condition for which the wing was designed,

$$C_{D_i} = \frac{C_L^2}{\pi R_A} \quad (9.3.16)$$

The ratio of induced drag coefficient to the square of the lift coefficient, for a unswept elliptic wing, is also plotted in Fig. 9.3.4 as a function of aspect ratio.

Figure 9.3.4 displays the two primary differences between the properties of a finite wing and those of the airfoil section from which the wing is generated. As previously discussed, a finite wing generates induced drag which is not produced by the airfoil section. In addition, the finite wing has a lift slope that is always less than the section lift slope of the wing's airfoil. The results shown in Fig. 9.3.4 are for a wing with an elliptic planform shape. At a given angle of attack, the elliptic wing produces less induced drag and more lift than any other wing of the same aspect ratio. The planform shape affects both the induced drag and the lift slope of a finite wing. However, as we shall see in the next section, the effect of planform shape is small compared to the effect of aspect ratio. Both the production of induced drag and the decrease in lift slope are a direct result of downwash. The downwash decreases the effective angle of attack, reducing the lift and tilting the lift vector back to produce the induced drag.

As shown in Fig. 9.3.4, for low aspect ratio, Eq. (9.3.14) predicts a very substantial decrease in the lift slope of a finite wing over that of the airfoil section. Here again, the classic wing data obtained by Prandtl (1921) also confirmed this relationship between lift slope and aspect ratio.

While an elliptic wing will minimize induced drag, maximize lift slope, and minimize the downwash induced on the wing itself, the elliptic planform shape does not necessarily minimize the downwash induced by the finite wing on other surfaces of the aircraft. The downwash induced near an elliptic wing can be computed from Eq. (9.2.26) with all of the Fourier coefficients except B_1 set to zero. This gives

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{\cos(\phi)[\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{\bar{x} \cos(\phi)[\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin^2(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.3.17)$$

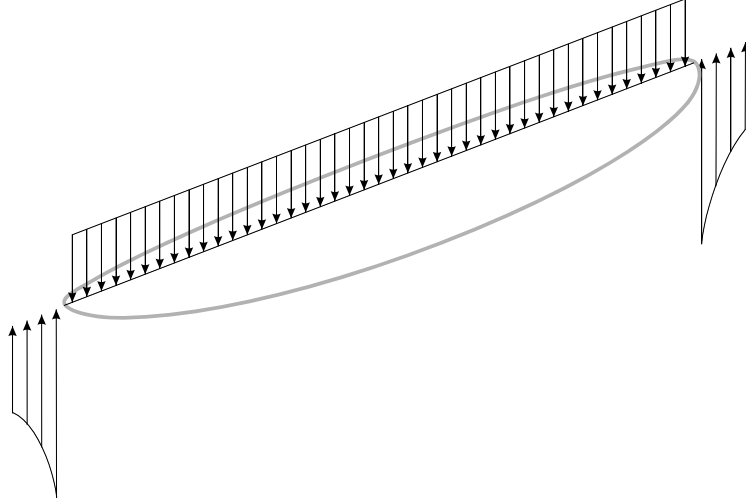


Figure 9.3.5. The downwash along the axis of an unswept elliptic wing with an aspect ratio of 6.

The downwash along the wing axis is,

$$V_y(0,0,\bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \left\{ \frac{1}{\pi} \int_0^\pi \frac{\cos(\phi)}{[\bar{z} + \cos(\phi)]} d\phi \right\} = -\frac{V_\infty C_L}{\pi R_A} \begin{cases} \left(1 + \frac{\bar{z}}{\sqrt{\bar{z}^2 - 1}} \right), & \bar{z} < -1 \\ 1, & \bar{z}^2 < 1 \\ \left(1 - \frac{\bar{z}}{\sqrt{\bar{z}^2 - 1}} \right), & \bar{z} > 1 \end{cases} \quad (9.3.18)$$

From Eq. (9.3.18), we see that an elliptic wing will result in downwash that is constant over the span of the wing itself, as is shown in Fig. 9.3.5. If the downwash is constant, the induced angle of attack, the effective angle of attack, and the local section lift coefficient will also be constant over the span of the wing. Using Eqs. (9.3.14) and (9.3.18) in Eq. (9.1.18) we have

$$\alpha_i = -\frac{V_y(0,0,z)}{V_\infty} = \frac{C_L}{\pi R_A} = B_1 = \frac{\tilde{C}_{L,\alpha}(\alpha - \alpha_{L0})}{\pi R_A + \tilde{C}_{L,\alpha}} \quad (9.3.19)$$

Since all Fourier coefficients except B_1 are zero, from Eq. (9.2.45) we see that the wingtip vortex strength factor, κ_v , is 1.0 for an elliptic wing. From Eq. (9.2.50), the wingtip vortex span factor, κ_b , is

$$\kappa_b = b'/b = \pi/4 \quad (9.3.20)$$

Thus, from Eq. (9.2.44), the downwash a few chord lengths or more down-stream from an elliptic wing is estimated as

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi^2 R_A} \left\{ \frac{\frac{\pi}{4} - \bar{z}}{\bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2}} \right] \right. \\ \left. + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[\frac{\frac{\pi}{4} - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2}} + \frac{\frac{\pi}{4} + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2}} \right] + \frac{\frac{\pi}{4} + \bar{z}}{\bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2}} \right] \right\} \quad (9.3.21)$$

In **summary**, the finite elliptic wing with no sweep, no dihedral, no geometric twist, and no aerodynamic twist results in the minimum induced drag for a given lift coefficient and aspect ratio.

The **chord length** for an elliptic wing varies with the spanwise coordinate according to,

$$c(z) = \frac{4b}{\pi R_A} \sqrt{1 - (2z/b)^2} \quad (9.3.12)$$

The **circulation distribution** for an elliptic wing is also elliptic,

$$\Gamma(z) = \frac{2bV_\infty \tilde{C}_{L,\alpha}(\alpha - \alpha_{L=0})}{\pi R_A + \tilde{C}_{L,\alpha}} \sqrt{1 - (2z/b)^2} \quad (9.3.13)$$

The **lift coefficient and induced drag coefficient** for an elliptic wing are,

$$C_L = C_{L,\alpha}(\alpha - \alpha_{L=0}); \quad C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{1 + \tilde{C}_{L,\alpha}/\pi R_A} \quad (9.3.14)$$

$$C_{D_i} = \frac{C_L^2}{\pi R_A} \quad (9.3.16)$$

The **downwash induced near an elliptic wing** is,

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi R_A} \frac{1}{\pi} \int_{\phi=0}^{\pi} \left\{ \frac{\cos(\phi)[\bar{z} + \cos(\phi)]}{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2} + \frac{\bar{x} \cos(\phi)[\bar{z} + \cos(\phi)]}{\{\bar{y}^2 + [\bar{z} + \cos(\phi)]^2\} \{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{1/2}} + \frac{\bar{x} \sin^2(\phi)}{\{\bar{x}^2 + \bar{y}^2 + [\bar{z} + \cos(\phi)]^2\}^{3/2}} \right\} d\phi \quad (9.3.17)$$

and the **downwash a few chord lengths or more downstream from an elliptic wing** is

$$V_y(\bar{x}, \bar{y}, \bar{z}) = -\frac{V_\infty C_L}{\pi^2 R_A} \left\{ \frac{\frac{\pi}{4} - \bar{z}}{\bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2}} \right] + \frac{\bar{x}}{\bar{x}^2 + \bar{y}^2} \left[\frac{\frac{\pi}{4} - \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} - \bar{z})^2}} + \frac{\frac{\pi}{4} + \bar{z}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2}} \right] + \frac{\frac{\pi}{4} + \bar{z}}{\bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2} \left[1 + \frac{\bar{x}}{\sqrt{\bar{x}^2 + \bar{y}^2 + (\frac{\pi}{4} + \bar{z})^2}} \right] \right\} \quad (9.3.21)$$

Both the downwash and the induced angle of attack for an elliptic wing remain constant over the span of the wing itself.